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| <b>Risk Assessment Record</b>                                                                            |                                                | Project: Soapamorph                         |                                       |                                    |                                                    | <i>To be assessed yearly</i> |  |
|                                                                                                          |                                                |                                             |                                       |                                    |                                                    |                              |  |
| <b>Risk Assessed By:</b>                                                                                 | Rachel Kirkham                                 |                                             |                                       |                                    |                                                    |                              |  |
| <b>Name:</b>                                                                                             | Group A                                        | <b>Position:</b>                            | Student                               | <b>Location:</b>                   | Mellor Building.                                   |                              |  |
| <b>Date:</b>                                                                                             | 22/02/2026                                     | <b>Level:</b>                               | 6                                     | <b>Department:</b>                 | Department of Computing                            |                              |  |
|                                                                                                          |                                                |                                             |                                       |                                    |                                                    |                              |  |
| <b>Risk Matrix – Use this to determine level of risk for each hazard i.e. ‘How bad &amp; How likely’</b> |                                                |                                             |                                       |                                    |                                                    |                              |  |
| <b>Severity of Harm</b>                                                                                  | <b>(1) Rare</b><br>(e.g. once every few years) | <b>(2) Unlikely</b><br>(e.g. Once per year) | <b>(3) Possible</b><br>(e.g. Monthly) | <b>(4) Likely</b><br>(e.g. Weekly) | <b>(5) Very Likely</b><br>(e.g. daily or frequent) |                              |  |
| <b>(1) Negligible</b><br>(e.g. Slight Bruising)                                                          | Low<br>(1)                                     | Low<br>(2)                                  | Low<br>(3)                            | Medium<br>(4)                      | Medium<br>(5)                                      |                              |  |
| <b>(2) Minor</b><br>(e.g. Small cuts, deep bruising)                                                     | Low<br>(2)                                     | Low<br>(4)                                  | Medium<br>(6)                         | Medium<br>(8)                      | Medium<br>(10)                                     |                              |  |
| <b>(3) Moderate</b><br>(e.g. Deep cuts, Torn muscles)                                                    | Low<br>(3)                                     | Medium<br>(6)                               | Medium<br>(9)                         | High<br>(12)                       | High<br>(15)                                       |                              |  |
| <b>(4) Severe</b><br>(e.g. Fracture, Loss of consciousness)                                              | Medium<br>(4)                                  | Medium<br>(8)                               | High<br>(12)                          | High<br>(16)                       | Extremely High<br>(20)                             |                              |  |
| <b>(5) Very Severe</b><br>(e.g. Death,                                                                   | Medium<br>(5)                                  | High<br>(10)                                | High<br>(15)                          | Extremely High<br>(20)             | Extremely High<br>(25)                             |                              |  |

| Permanent disability/disease) |                                                                                                                                    |                                                                                                                                                                                                                                                           |                  |                |                                     |                |                                                                                                                                                                                                           |                                                                               |
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| Legend:                       | Low (1-4)                                                                                                                          |                                                                                                                                                                                                                                                           | Medium (4-10)    |                | High (11-15)                        |                | Extreme (16-25)                                                                                                                                                                                           |                                                                               |
| Hazard List – List below.     |                                                                                                                                    |                                                                                                                                                                                                                                                           |                  |                |                                     |                |                                                                                                                                                                                                           |                                                                               |
| Hazards identified            | Risk                                                                                                                               | Potential injuries/ damage                                                                                                                                                                                                                                | Likelihood (1-5) | Severity (1-5) | Risk Rating (Likelihood x Severity) | Who is at risk | Additional measures/ Precautions made?                                                                                                                                                                    | Who is responsible for action?                                                |
| Electrical or battery fault   | <ul style="list-style-type: none"><li>Chemical Leak/ burns</li><li>Fire/ overheating</li><li>Electric shock</li><li>Fire</li></ul> | <ul style="list-style-type: none"><li>Chemical burns</li><li>Laboratory damage</li><li>Electric shock</li><li>Shock</li><li>Burns</li></ul>                                                                                                               | 2                | 4              | 8 (Medium)                          | Team members   | <ul style="list-style-type: none"><li>All electrical equipment will have been PAT tested.</li><li>Pre-use inspection.</li><li>Switch on main power.</li></ul>                                             | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |
| Animatronic Malfunction       | <ul style="list-style-type: none"><li>Unexpected or uncontrolled movement</li><li>Minor injury</li></ul>                           | <ul style="list-style-type: none"><li>Bruising</li></ul>                                                                                                                                                                                                  | 2                | 1              | 2 (Low)                             | Team members   | <ul style="list-style-type: none"><li>Switch on main power.</li><li>Plenty of space to stand away from project.</li></ul>                                                                                 | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |
| Obstacles / Cluttered Area    | <ul style="list-style-type: none"><li>Trip/slip hazard</li><li>Collison</li><li>Blockage of exits</li></ul>                        | <ul style="list-style-type: none"><li>Slip/ Trip</li><li>Injury to back</li><li>Sprain/ break</li><li>Head injury</li><li>Smoke inhalation</li><li>Blocked in fire</li><li>Minor injuries from cluttered workspace (such as dropping onto feet)</li></ul> | 2                | 4              | 8 (Medium)                          | Team members   | <ul style="list-style-type: none"><li>There will be plenty of manoeuvrable space.</li><li>Exits and walkways kept clear.</li><li>Storage locations for belongings.</li><li>Workplace kept tidy.</li></ul> | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |
| Objects falling               | <ul style="list-style-type: none"><li>Heavy items</li></ul>                                                                        | <ul style="list-style-type: none"><li>Breakages</li></ul>                                                                                                                                                                                                 |                  |                |                                     | Team           | <ul style="list-style-type: none"><li>Tidy workspace.</li></ul>                                                                                                                                           | Rachel                                                                        |

|                         |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                 |   |   |                |              |                                                                                                                                                                                                                                                                                                                                                                                               |                                                                               |
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|                         | <ul style="list-style-type: none"> <li>falling Tools falling.</li> </ul>                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Bruising</li> <li>Fracture</li> </ul>                                                                                                                                                    | 1 | 4 | 4<br>(Low)     | members      | <ul style="list-style-type: none"> <li>No cluttered surfaces.</li> <li>Storage area for belongings.</li> <li>Appropriate use of PPE - steel toe caps.</li> </ul>                                                                                                                                                                                                                              | Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller        |
| Cutting, welding metal. | <ul style="list-style-type: none"> <li>Burns.</li> <li>Sparks.</li> <li>Foreign objects.</li> <li>Contact with equipment.</li> <li>Bright/piercing lights.</li> <li>Snagged clothing.</li> <li>Untidy workplace.</li> <li>Falling objects.</li> <li>Damaged equipment.</li> </ul> | <ul style="list-style-type: none"> <li>Blindness.</li> <li>Eye injuries.</li> <li>Severe burns.</li> <li>Small cuts.</li> <li>Deep cuts.</li> <li>Injured limbs.</li> <li>Damage to feet.</li> <li>Long term damage.</li> </ul> | 2 | 5 | 10<br>(Medium) | Team members | <ul style="list-style-type: none"> <li>Wear appropriate PPE (Gloves, Welding mask, Goggles)</li> <li>Wear appropriate tight clothing - tight fitted.</li> <li>Wear steel toe cap safety boots.</li> <li>Keep workstation tidy and all exits/ walkways tidy.</li> <li>All equipment checked before and after use for wear and tear.</li> <li>All equipment checked for PAT testing.</li> </ul> | Rachel Kirkham.                                                               |
| Soldering.              | <ul style="list-style-type: none"> <li>Contact with equipment.</li> <li>Fire.</li> <li>Untidy workplace.</li> <li>Falling objects.</li> </ul>                                                                                                                                     | <ul style="list-style-type: none"> <li>Eye injuries.</li> <li>Severe burns.</li> <li>Lung disease/ injury.</li> <li>Electrocution.</li> <li>Shock.</li> </ul>                                                                   | 2 | 5 | 10<br>(Medium) | Team members | <ul style="list-style-type: none"> <li>Wearing of appropriate PPE (goggles and steel toe caps).</li> <li>Wearing</li> </ul>                                                                                                                                                                                                                                                                   | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |

|                   |                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                           |   |   |               |              |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                               |
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|                   | <ul style="list-style-type: none"> <li>Fumes.</li> <li>Foreign objects/ Objects in eye.</li> <li>Damaged equipment.</li> </ul>                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                           |   |   |               |              | <ul style="list-style-type: none"> <li>appropriate tight fitted clothing.</li> <li>Stand up during soldering process to limit risk of solder dripping onto legs.</li> <li>All equipment checked for wear and tear before and after use.</li> <li>All equipment checked for PAT testing.</li> </ul>                                                                                                          |                                                                               |
| Silicone moulding | <ul style="list-style-type: none"> <li>Toxic fumes.</li> <li>Chemical exposure.</li> <li>Poor ventilation.</li> <li>Heat resistance failures.</li> <li>Incomplete curing.</li> <li>Incorrect manual handling.</li> <li>Sharp tools and equipment.</li> <li>Spills.</li> <li>Cluttered work area.</li> </ul> | <ul style="list-style-type: none"> <li>Skin irritation.</li> <li>Chemical burns.</li> <li>Allergic reactions.</li> <li>Respiratory irritation.</li> <li>Headaches.</li> <li>Dizziness.</li> <li>Nausea.</li> <li>Skin blistering.</li> <li>Minor to Severe burns.</li> <li>Muscle or back strain.</li> <li>Cuts/punctures.</li> <li>Slips and falls.</li> <li>Falling objects.</li> </ul> | 2 | 4 | 8<br>(Medium) | Team members | <ul style="list-style-type: none"> <li>Wear appropriate PPE, gloves, goggles, lab coat, steel toe cap boots etc....</li> <li>Keep a well-ventilated area.</li> <li>Heat sources supervised.</li> <li>Keep workspace tidy and clean any spillages.</li> <li>Wear gloves for sharp tool exposure.</li> <li>Follow COSHH manufacturer data sheet.</li> <li>Wash hands before and after (don't touch</li> </ul> | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |

|               |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                 |   |   |            |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                               |
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|               |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                 |   |   |            |              | face during process).                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                               |
| Laser cutting | <ul style="list-style-type: none"> <li>• Fire.</li> <li>• Poor ventilation.</li> <li>• Exposure to laser lighting.</li> <li>• Moving parts/mechanical injury/malfunctions.</li> <li>• Damages to equipment.</li> <li>• Damages to wiring or power source.</li> <li>• Sharp edges/broken material handling.</li> </ul> | <ul style="list-style-type: none"> <li>• Smoke/ toxic fume inhalation.</li> <li>• Eye strain to permanent damage/blindness.</li> <li>• Cuts/punctures.</li> <li>• Electrical shock.</li> <li>• Bruising.</li> <li>• Minor to major muscle injury.</li> <li>• Respiratory irritation/long term damage.</li> <li>• Headaches.</li> <li>• Dizziness.</li> <li>• Nausea.</li> </ul> | 1 | 4 | 4<br>(Low) | Team members | <ul style="list-style-type: none"> <li>• Wear appropriate PPE, glasses if needed.</li> <li>• Make sure ventilation on machine works.</li> <li>• Check machine for PAT testing.</li> <li>• Check machine for external and internal damages.</li> <li>• Check wiring to power socket.</li> <li>• Make sure emergency button is easily nearby.</li> <li>• Do not get material out of machine for 20 seconds to 1 minute to allow cooling and ventilation.</li> </ul> | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |
| 3D printing   | <ul style="list-style-type: none"> <li>• Plastic fumes.</li> <li>• Fire.</li> <li>• Poor ventilation.</li> <li>• Heat buildup.</li> <li>• Malfunctions.</li> <li>• Exposed/damaged wiring.</li> <li>• Damages to machine present.</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>• Respiratory irritation/long term damage.</li> <li>• Headaches.</li> <li>• Dizziness.</li> <li>• Nausea.</li> <li>• Electrical shock.</li> <li>• Smoke/ toxic fume inhalation.</li> </ul>                                                                                                                                               | 1 | 4 | 4<br>(Low) | Team members | <ul style="list-style-type: none"> <li>• Check machine ventilation.</li> <li>• Check machine has been PAT tested.</li> <li>• Check internal and external parts of machine for signs of damage before</li> </ul>                                                                                                                                                                                                                                                   | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |

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|                     |                                                                                                            |                                                                                                  |   |   |            |              | <ul style="list-style-type: none"> <li>• Check wiring and main power connection for signs of damage.</li> </ul> |                                                                               |
| Fusion/blender      | <ul style="list-style-type: none"> <li>• Screen strain.</li> <li>• Incorrect posture long term.</li> </ul> | <ul style="list-style-type: none"> <li>• Eye strain.</li> <li>• Back / muscle strain.</li> </ul> | 2 | 2 | 4<br>(Low) | Team members | <ul style="list-style-type: none"> <li>• Take consistent breaks throughout.</li> </ul>                          | Rachel Kirkham<br>Chris Mead<br>Hannah Arnold<br>Max McGill<br>Balraj Bhuller |
|                     |                                                                                                            |                                                                                                  |   |   |            |              |                                                                                                                 |                                                                               |
| Assessor Signature: | R.Kirkham                                                                                                  |                                                                                                  |   |   |            |              |                                                                                                                 |                                                                               |
| Date:               | 22/02/2026                                                                                                 |                                                                                                  |   |   |            |              |                                                                                                                 |                                                                               |